ADVANCED IOT WEATHER STATION

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AGENDA

Part I - Product Introduction

1. Key Product Features
2. Product Specifications

Part II - Technical Overview

1. Major Components USED
2. Severe Weather Alert Thresholds
3. System Block Diagram
4. Software Diagram
5. Major Implementation Methods
6. Serial Communications Protocol

Part III - Conclusion

1. Viability of the Product
2. Future Developments

KEY PRODUCT FEATURES

- Real-time monitoring for weather conditions
- Local display with a 16x2 LCD
- Local alerts via buzzer and LED
- SMS alerts for severe weather conditions
- Remote monitoring through the Blynk IoT platform
- Scalable design for additional sensors

PRODUCT SPECIFICATIONS

OPERATING SYSTEM: LINUX-BASED (RASPBERRY PI 3 MODEL A+)

POWER INPUT: 12V DC

PCB SIZE: 4.5 X 5.5 INCHES

WEIGHT: 1 KG

TEMPERATURE RANGE: 0°C TO 50°C

Real-time updates on the Blynk IoT platform

Display: 16x2 LCD for real-time weather data display

Humidity Range: 20% to 90% Relative Humidity (RH)

Wind Speed Range: 0 to 30 (m/s)

SMS Notifications: Sent when parameters exceed predefined thresholds

Local Alerts: Buzzer and red LED for instant warnings

Customizable Thresholds: User-defined thresholds for wind speed, temperature, and rainfall

MAJOR COMPONENTS USED

- RASPBERRY PI 3 MODEL A+ (CONTROLLER)
- SENSORS: DHT11 (TEMP/HUMIDITY), RAINDROP, WIND SPEED
- SIM7670C GSM MODULE
- 16X2 LCD DISPLAY
- BUZZER
- LM2576 VOLTAGE REGULATOR
- MCP3208 ADC
- MAX232 IC

SEVERE WEATHER ALERT THRESHOLDS

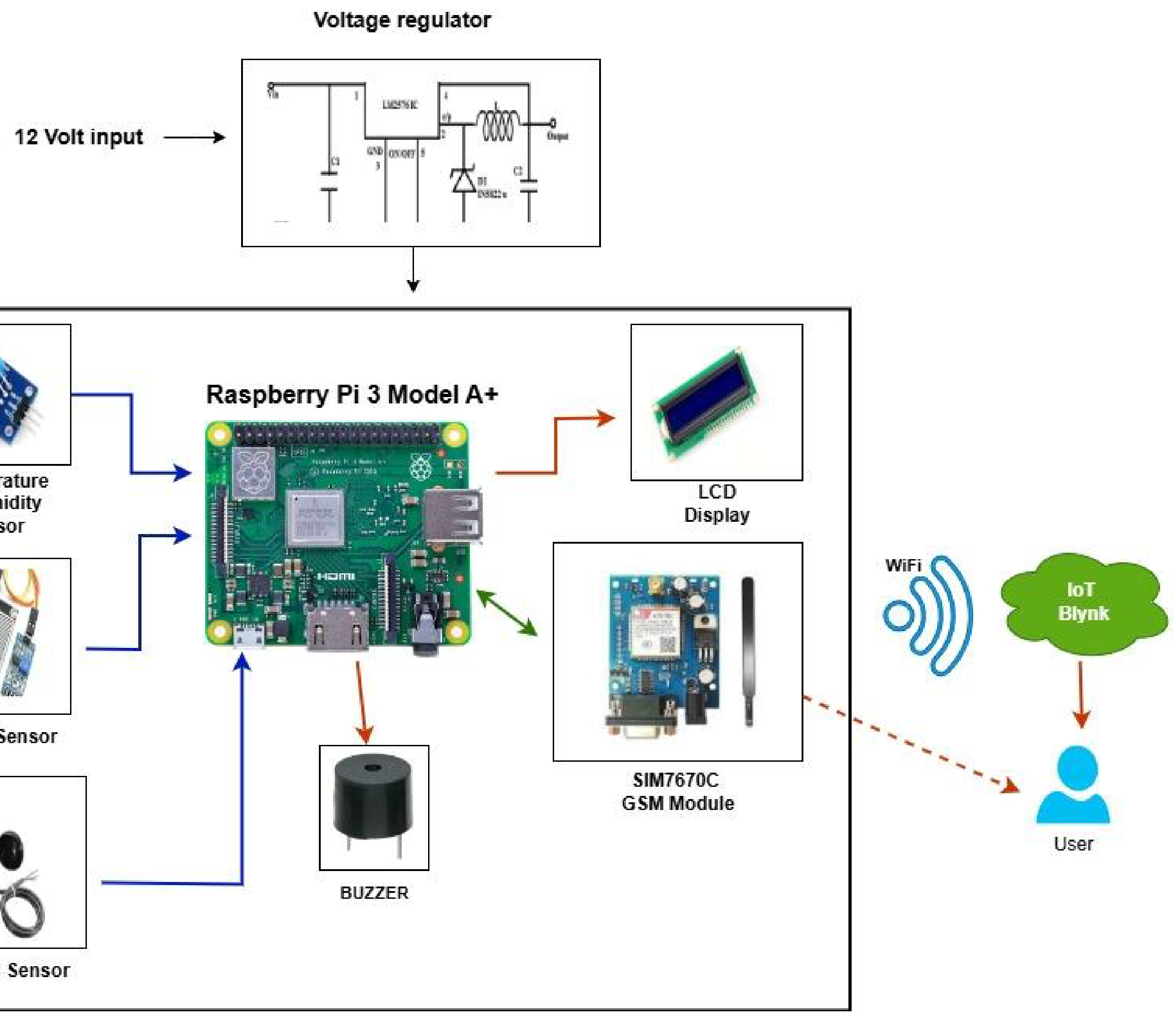
Heavy Rainfall: Rain level above 30mm

High Wind Speed: Wind speed above 25mph

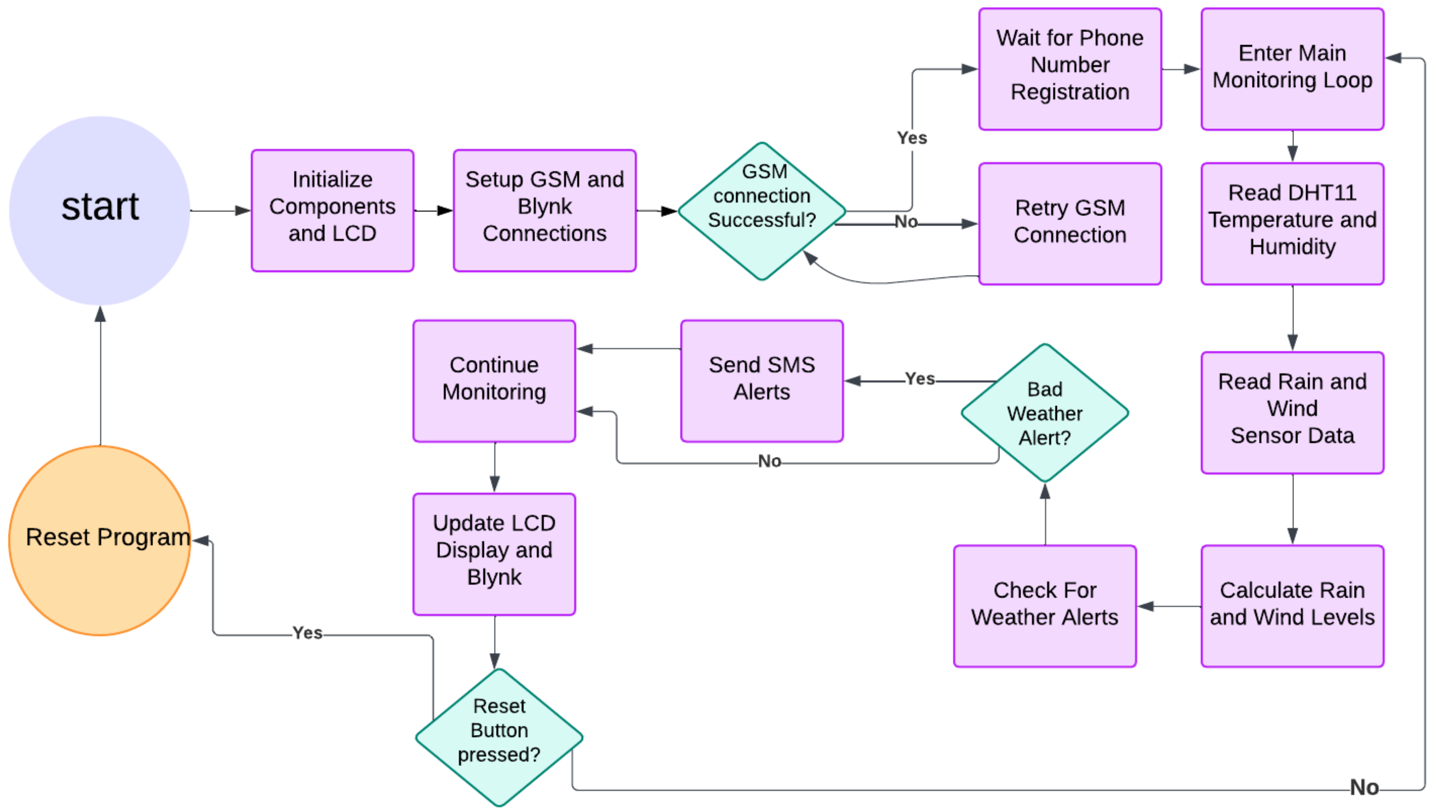
High Temperature: Temperature above 35°C

Low Temperature: Temperature below 5°C

SYSTEM BLOCK DIAGRAM



SOFTWARE DIAGRAM



MAJOR IMPLEMENTATION METHODS

1. **Software Implementation:**

- **Python Libraries:** Utilized key libraries such as:
  - **RPi.GPIO:** GPIO pin control for sensors and alerts.
  - **spidev:** SPI communication for MCP3208 ADC.
  - **adafruit_character_lcd:** LCD display management.
  - **adafruit_dht:** Data acquisition from the DHT11 sensor.
  - **serial:** Communication with the GSM module.
  - **requests:** Sending real-time updates to the Blynk platform.
- **Custom Functions:** Developed Python scripts for:
  - **Sensor Data Processing:** Capturing and converting environmental data.
  - **System Alerts:** Triggering SMS notifications and local alerts for severe weather conditions.

2. **Integration of Hardware Components:**

- Connected sensors (DHT11, raindrop sensor, wind speed sensor) to the Raspberry Pi GPIO and MCP3208 ADC module.
- MAX232 IC used for converting TTL to RS-232 levels for GSM module communication.

SERIAL COMMUNICATIONS PROTOCOL

1. **Purpose of Serial Communication**
   - Enables communication between the Raspberry Pi and GSM module for SMS alerts.

2. **Protocol Used**
   - **RS-232 Communication:**
     - Uses MAX232 IC to convert TTL (0–5V) signals to RS-232 levels (±25V).
   - Commands sent in ASCII text format over UART.

3. **Key AT Commands for GSM Module:**
   - **AT+CMGF=1:** Set SMS mode to text.
   - **AT+CMGS="phone_number":** Send SMS to a specified phone number.
   - **AT+CSQ:** Check signal strength.

4. **Packet Structure:**
   - Commands start with "AT", followed by parameters.
   - Responses received as ASCII strings (e.g., **OK**, **ERROR**).

4. **Error Handling:**
   - Commands retried on failure.
   - Signal strength and network registration checked periodically.

VIABILITY OF THE PRODUCT

- Cost-effective and scalable for diverse applications
- Suitable for agriculture, disaster management, and smart homes
- Provides critical real-time weather alerts and data visualization
- Customizable thresholds for temperature, wind speed, and rainfall alerts
- Remote monitoring ensures accessibility from any location
- Energy-efficient design with potential for solar power integration
- Enhances safety, productivity, and resource optimization

FUTURE DEVELOPMENTS

Addition of advanced sensors (pollution, snow and UV index)

Integration of solar power for energy efficiency

Development of a dedicated mobile application

Geolocation-based alerts for regional specificity

Support for LoRaWAN in rural areas

Voice assistant compatibility for smart home integration

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